GAS LEAK DETECTION SYSTEM

GLD V2

Intelligent, wireless multi-gas leak detection for refineries.



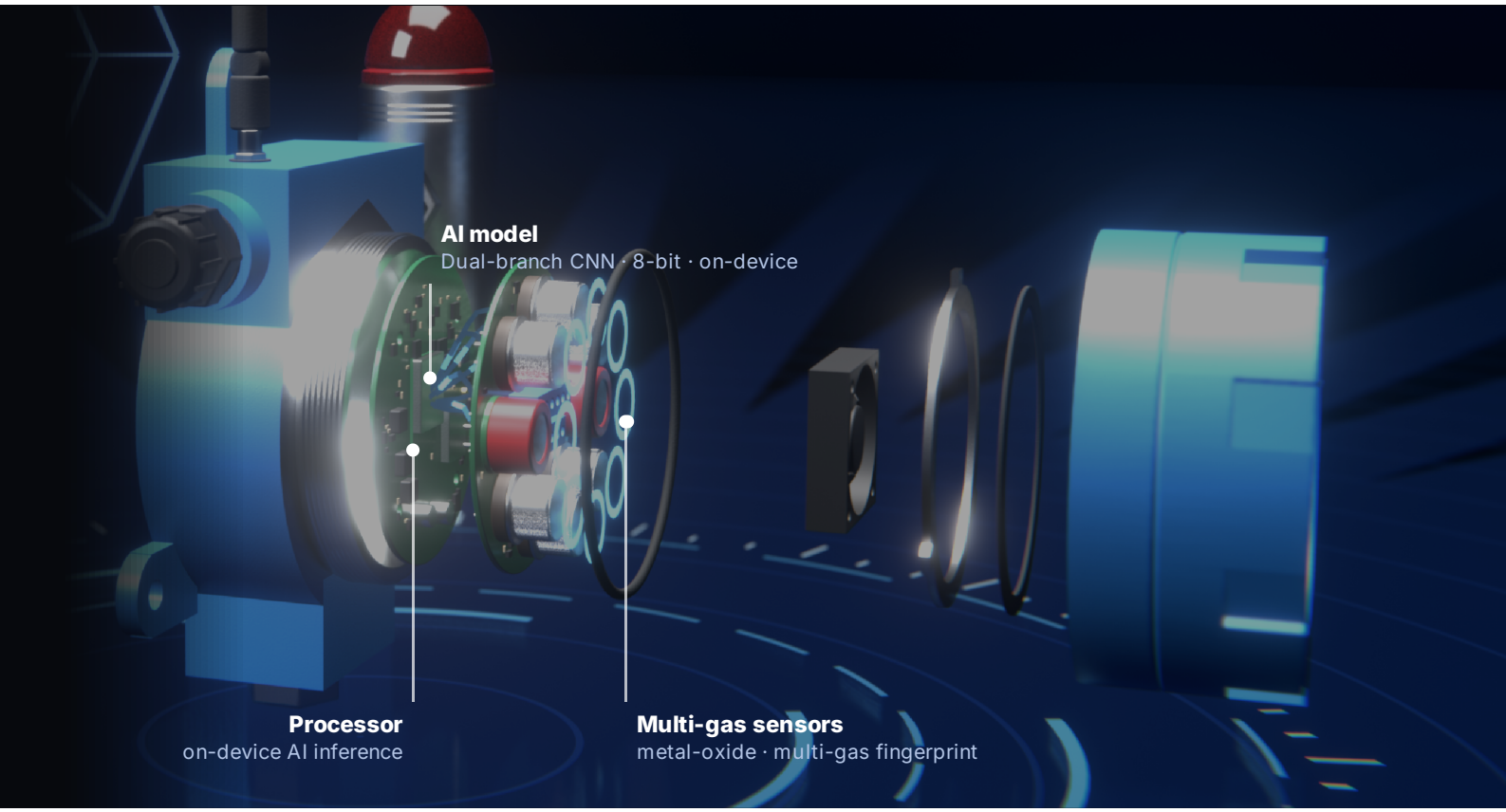
PT Pertamina Patra Niaga (Fungsi Reliability)
LAPI Ganesha Utama · Institut Teknologi Bandung

3D illustration

THE DEVICE

Multi-gas sensors. One device.

Each GLD reads the response pattern of multiple gas sensors, and on-device AI identifies the gas. No server round-trip, no waiting.



3D illustration of GLD V2, exploded view.

Gas fingerprint

Metal-oxide sensors form a distinct pattern for each gas.

Edge AI

A dual-branch 8-bit CNN runs directly on the microcontroller.

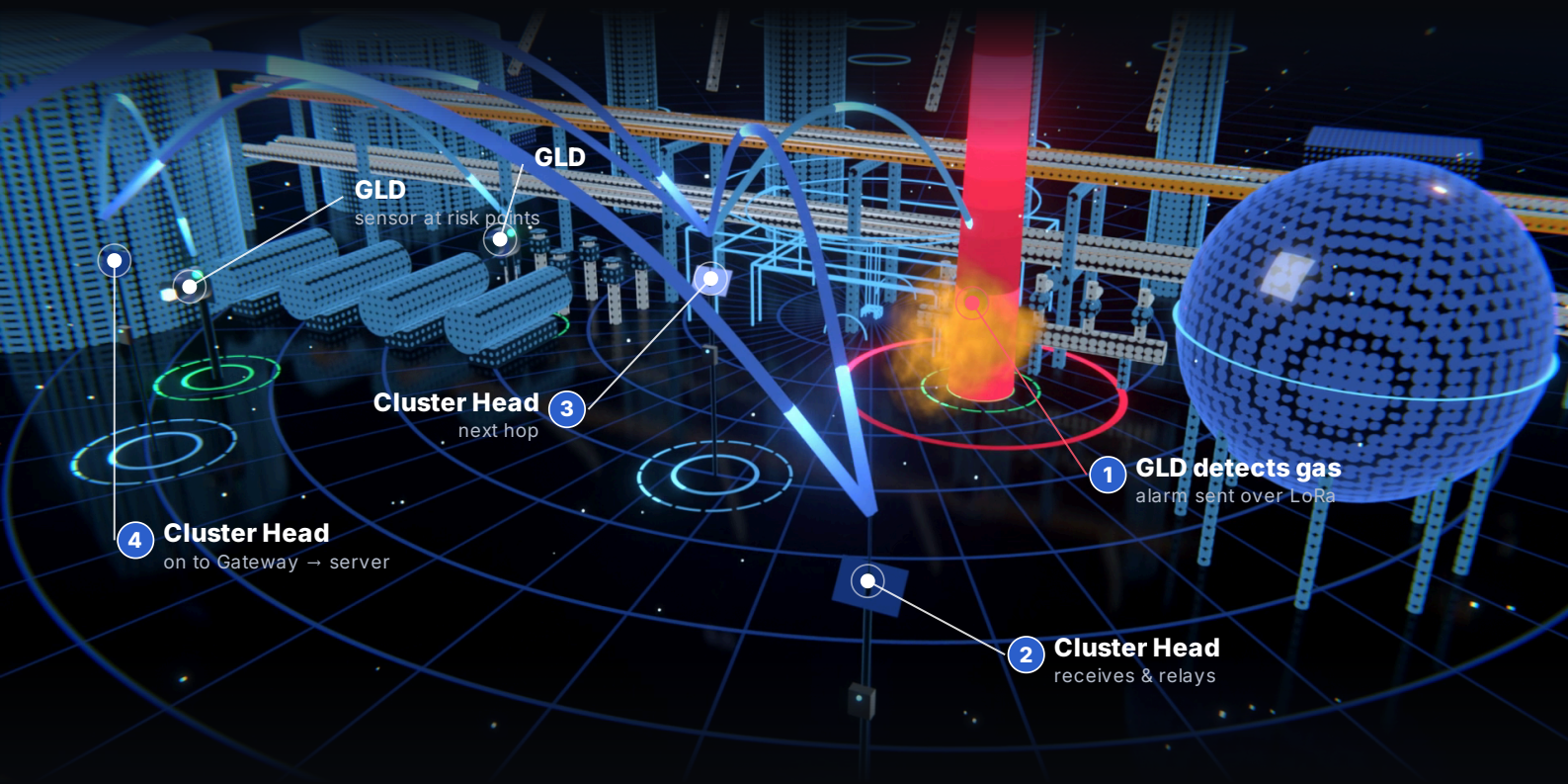
Local alarm

The on-device beacon lights up the moment gas is detected.

THE SYSTEM

Leak detected. Operators know instantly.

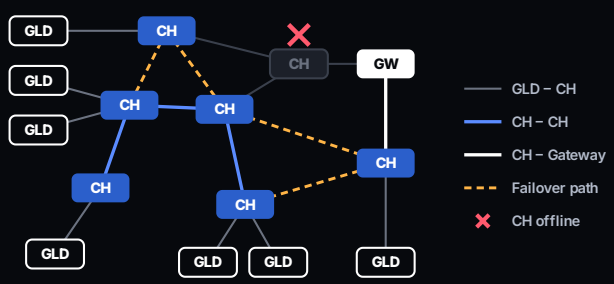
Data travels over a private LoRa mesh. No data cabling, no internet. If one path drops, the network reroutes itself.



GLD → Cluster Head → Gateway → Local server → Dashboard

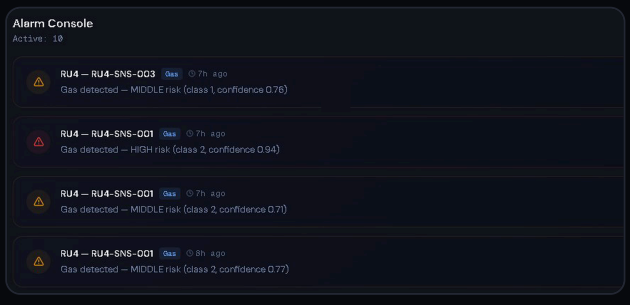
Self-healing network

If a Cluster Head goes offline, data is automatically rerouted through another path.



Instant push alarms

Alarms bypass the polling cycle; operators see location and risk level at once.



3D illustrations · dashboard shown with demonstration data.

PROVEN

From lab to refinery.

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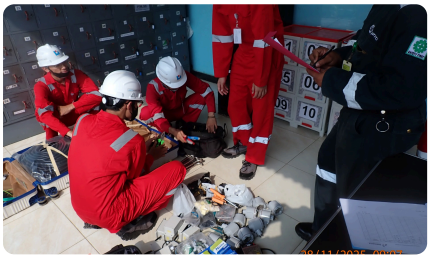
Factory Acceptance Test items passed · second-generation GLD, laboratory

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Cluster Heads in a multi-hop mesh, failover tested · ITB campus

2025

Field trial at RU VII Kasim, Nov–Dec · first-generation GLD



Equipment check with HSE personnel



Pole installation, witnessed by refinery staff



Installation on a solar-powered pole



GLD installed in the tank area



Data monitoring from the control room



Team coordination on site

Field trial at RU VII Kasim · November–December 2025 · first-generation GLD.

SPECIFICATIONS

Sensors	Multiple metal-oxide gas sensors
Intelligence	Dual-branch CNN, 8-bit, on-device
Communication	LoRa, private multi-hop mesh
Power supply	24 VDC
Validated gases	LPG, H_2
In development	CO_2 (data available) · CH_4, CO, NH_3, benzene (targets) · models trained on site

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Pilot: RU IV Cilacap